

INCS_810-Topics in Computer Security

Final Project: Tip Jar

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Abstract

This is a simple decentralized application (DApp) example that implements a "Tip Jar" smart contract. Anyone can send Ethereum to the contract as a tip. The contract automatically tracks the total tip amount and the number of tippers. The contract deployer (owner) can withdraw all accumulated tips.

Contract Summary

Key Variables:

- owner: The address of the contract owner.
- totalTips: The total amount of Ethereum received (in wei).
- totalTippers: The total number of tippers.

Key Functions:

- tip(): Send a tip to the contract (must be called with Ethereum).
- withdrawTips(): Can only be called by the owner to withdraw all tips.
- getContractBalance(): Check the current Ethereum balance of the contract.

Events:

- TipReceived: Triggered when someone sends a tip.
- TipsWithdrawn: Triggered when the owner withdraws the tips.

Compile/Run/Deploy

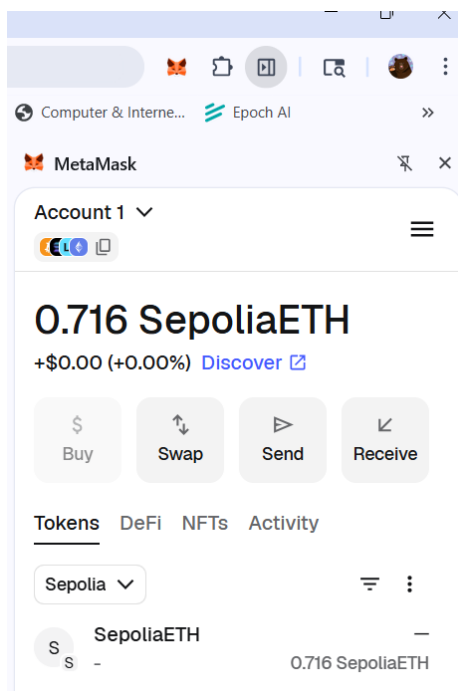
- Use Remix IDE
- Filename: TipJar.sol
- Testnet Used: Sepolia

Transactions

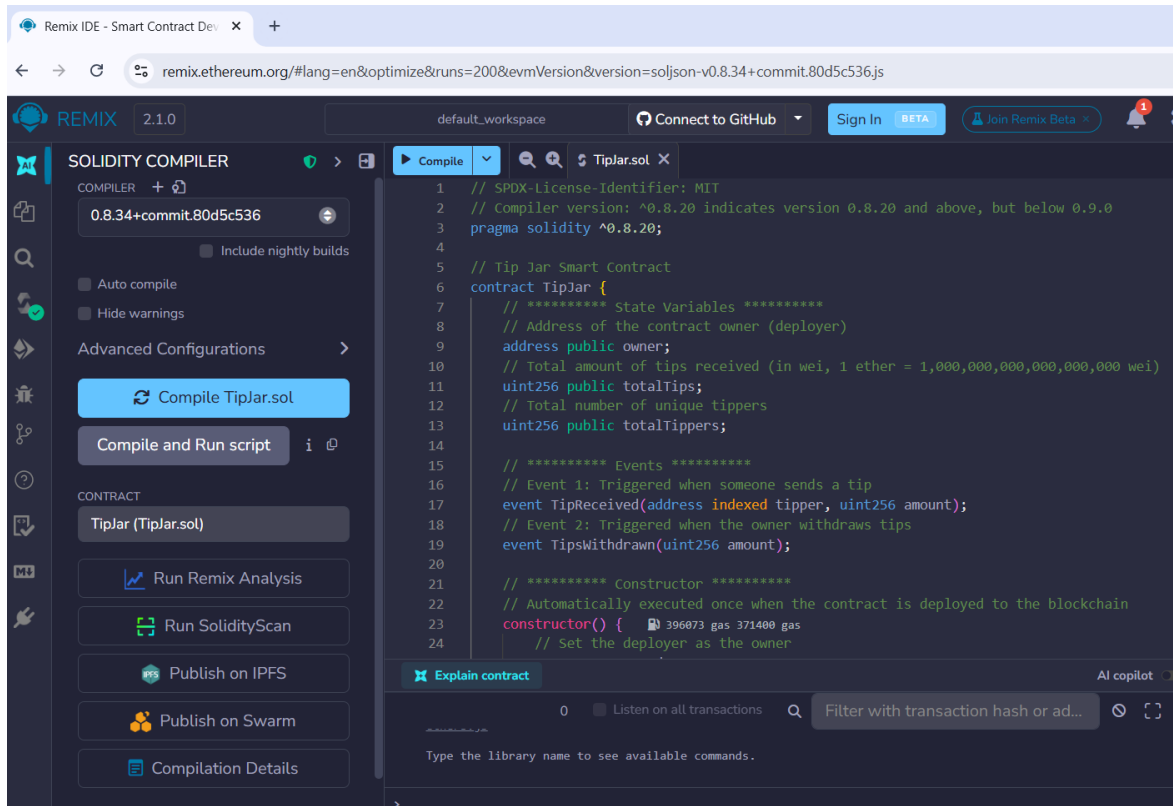
- Contract Address: **0xA8b470C5B8FcE7Cb8a82F9Fe6Cd4695883D4dBFd**
- Block Explorer Contract Link:
<https://sepolia.etherscan.io/address/0xA8b470C5B8FcE7Cb8a82F9Fe6Cd4695883D4dBFd>
- Deployment Transaction Hash:
0xc8b7b4951c32ea1fe1b0953cb30a39f7f99c980f9b66b1fe38a0563356c41a6a
- Example Interaction Transaction Hash (Tip):
0x6b93f5fac8dbee6be6d7b53642121b5f8b97ed610423270289f499f21e7bd1bd
- Example Interaction Transaction Hash (Withdrawal):
0x5a4e3d18b7de3fb062cb10d72290847e7503f9129219ba733ea727687c11b2a5

Walkthrough of Tip Jar Project

1. Setup MetaMask at Sepolia, shown the balance at the time.

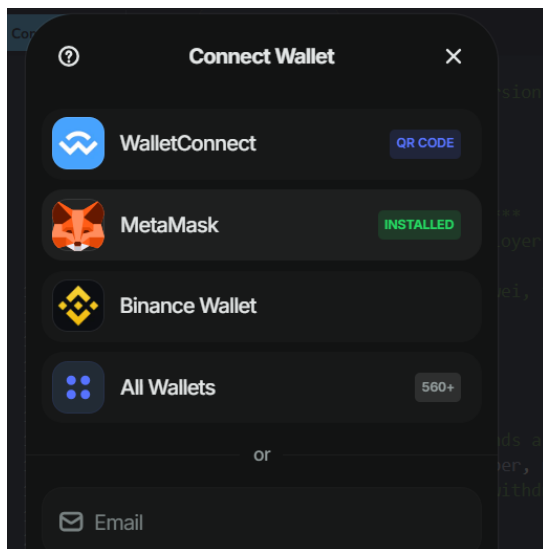


2. Run Remix and compile Tipjar.sol

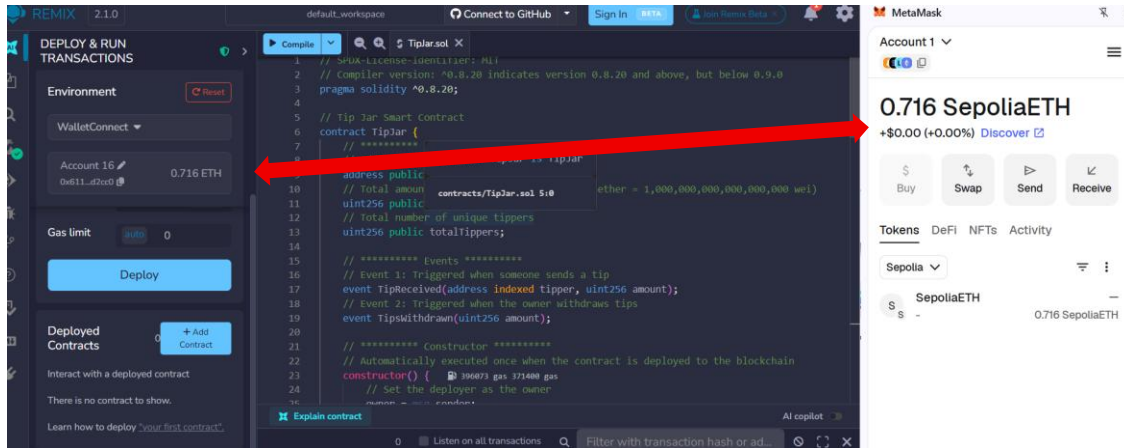


3. Deploy to Sepolia Testnet

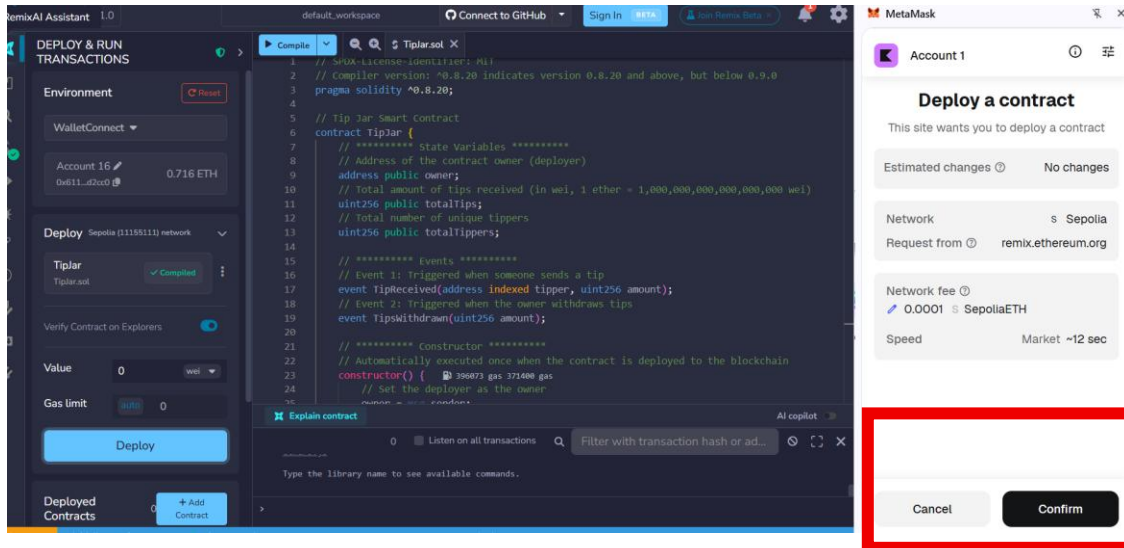
Browser Extension → MetaMask



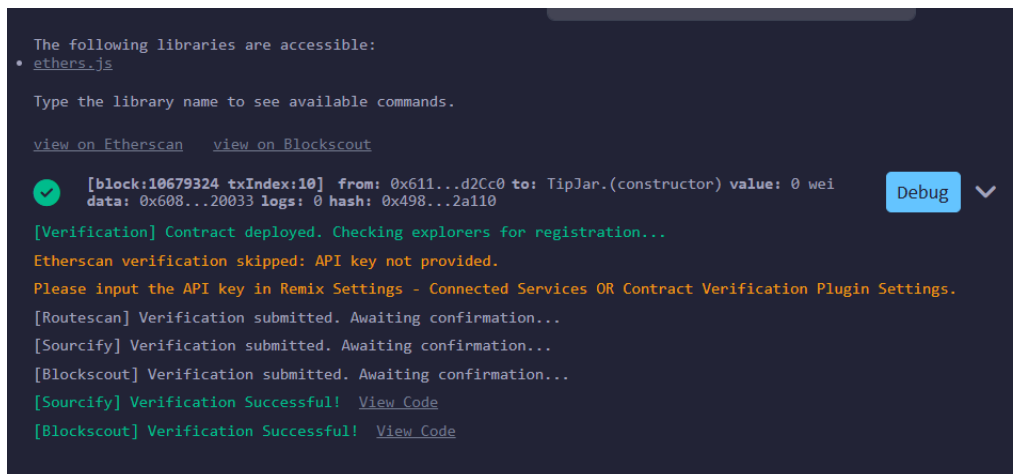
Connect to MetaMask. Remix will sync my wallet balance with MetaMask.



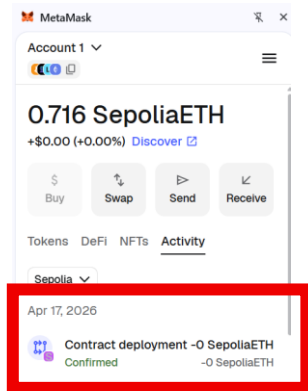
Deploy the contract and confirm at MetaMask.



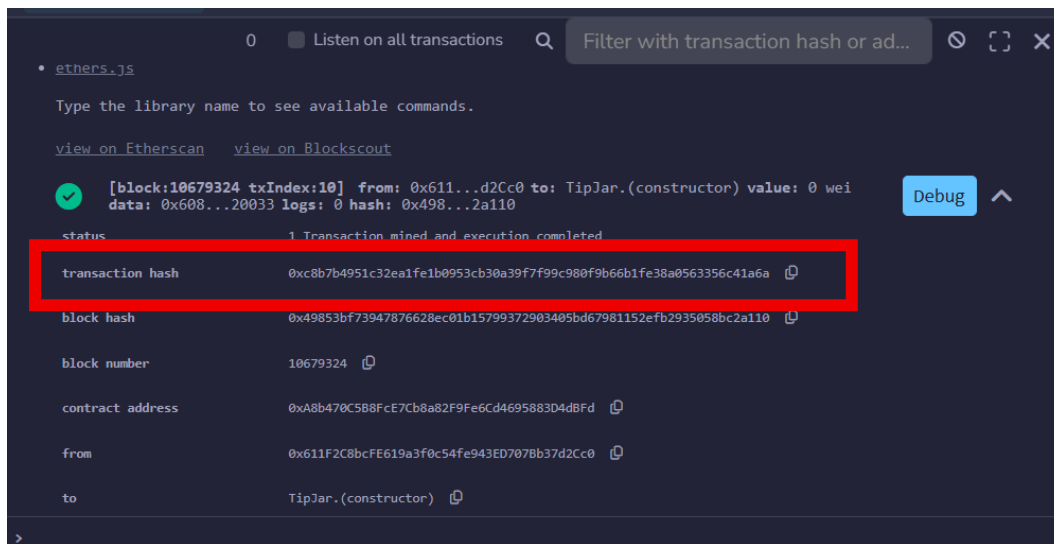
Deployed successfully.



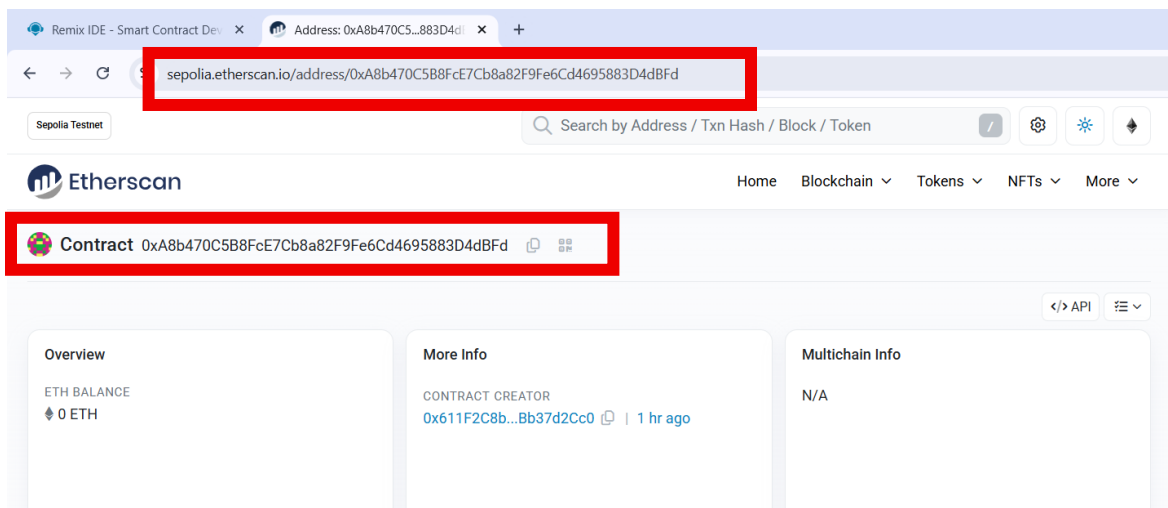
From MetaMask, it shows the deployment is successful.



Check the status on-chain. Copy the transaction hash:

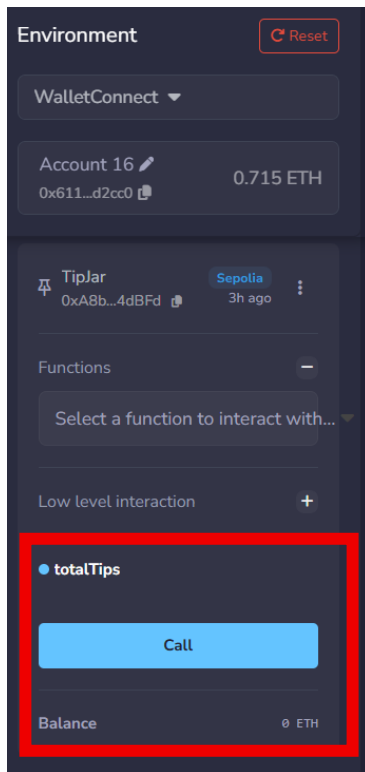


Look up the transaction at sepolia.etherscan.io, shown as below.

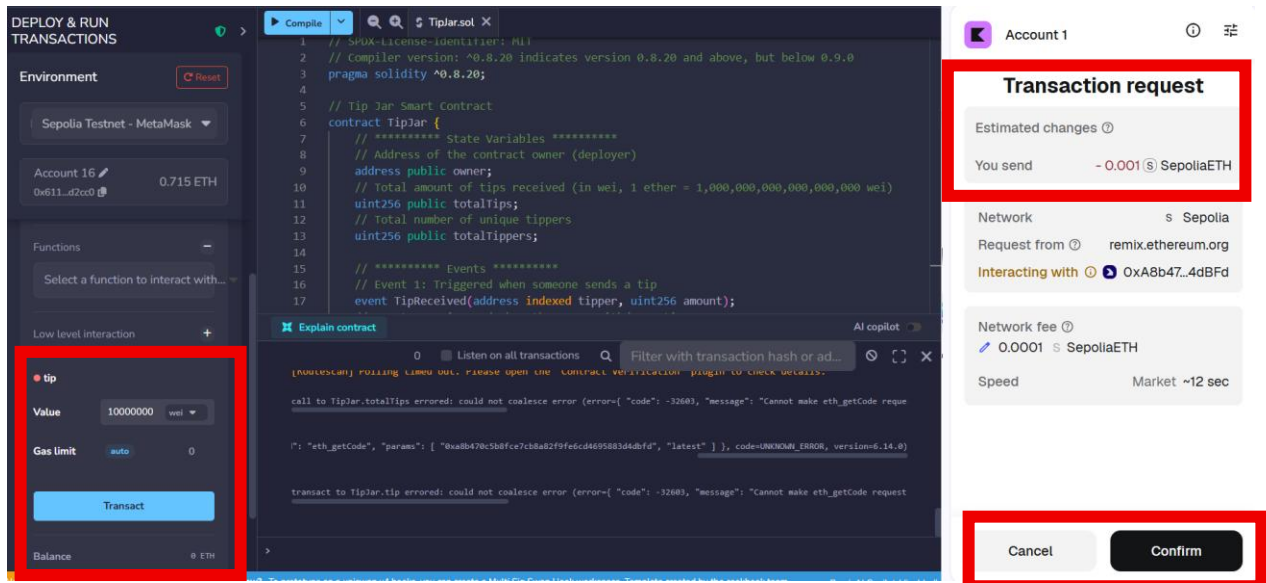


4. Test

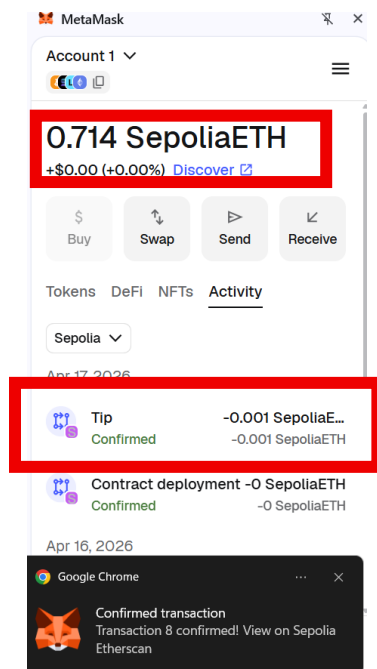
At the beginning, the Balance of totalTips is 0, because no tip transaction happens.



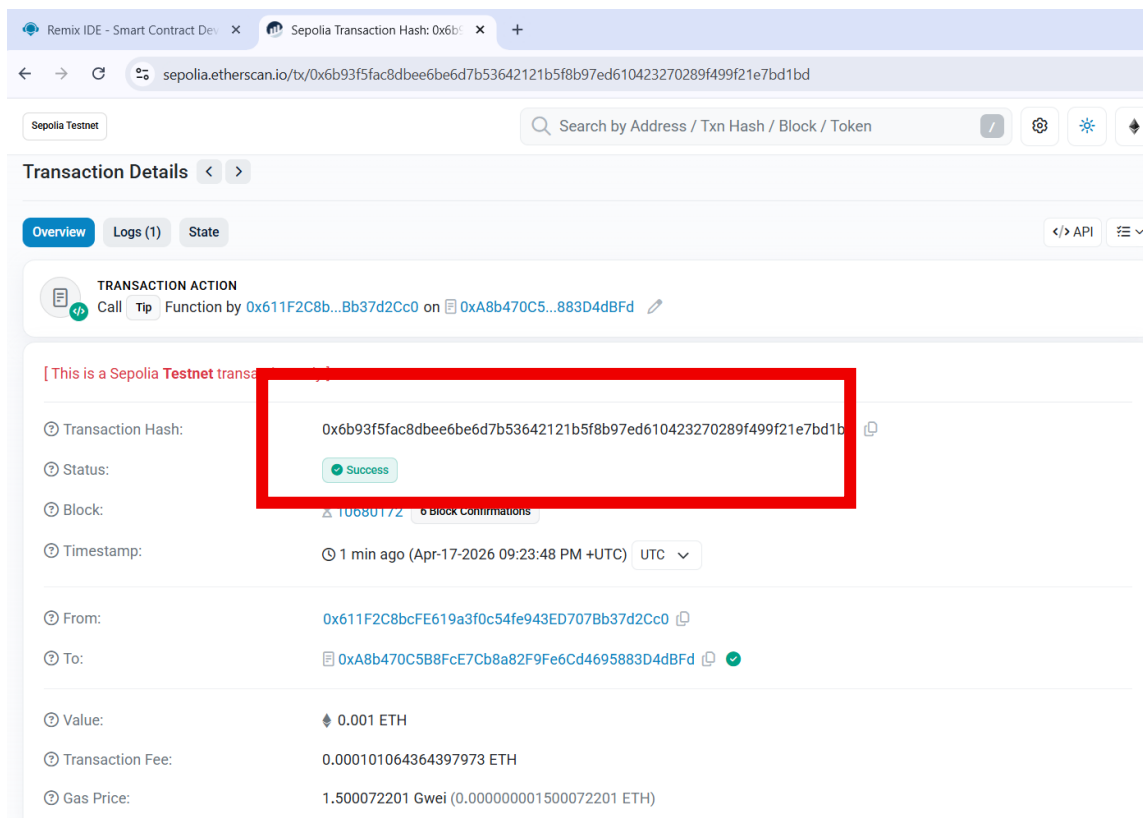
Then, I send 1×10^{15} wei as tips to the contract.



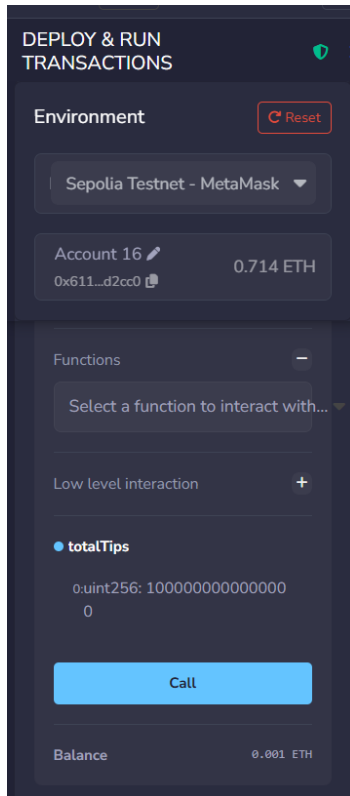
From my wallet at MetaMask, we can see the tip (0.001SepoliaETH) deducted.



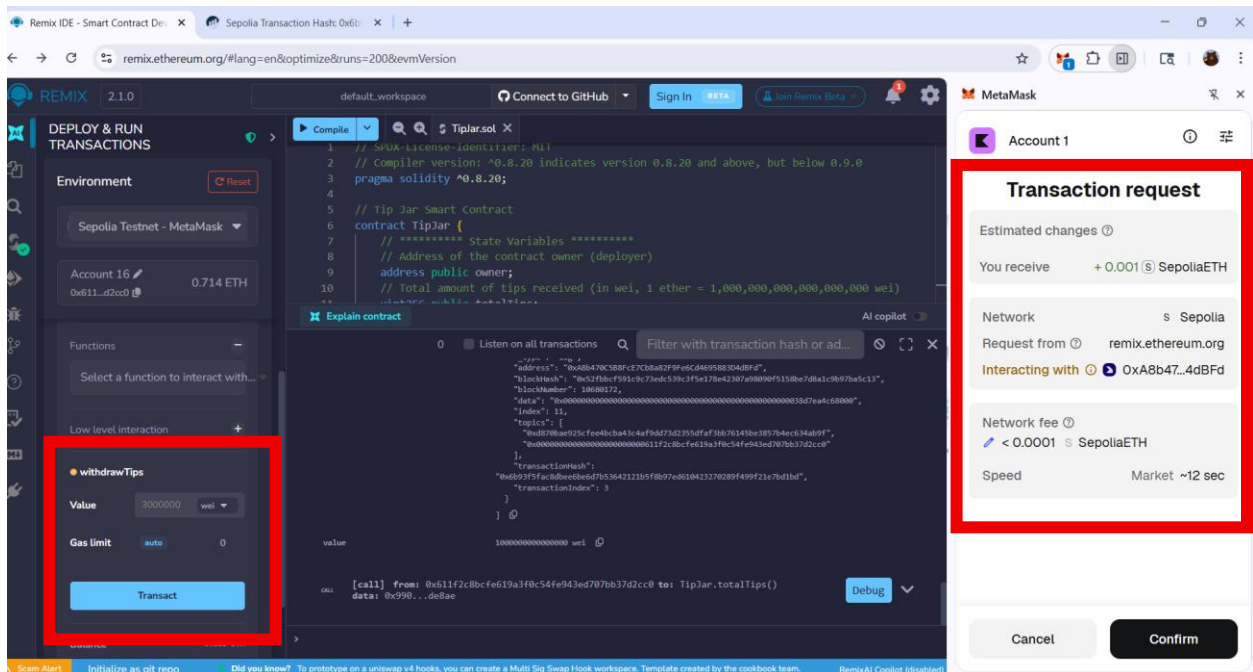
We can check the details of this Tip transaction on sepolia.etherscan.io.



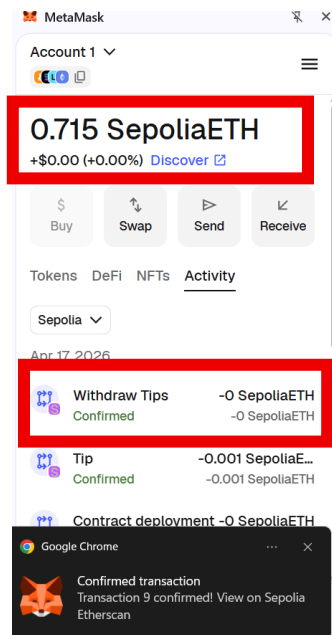
At Remix, I check the totalTips function, and it returns the balance 0.001 ETH.



Last, as the contract owner, I would like to withdraw all the tips I received.



This operation was approved and my wallet at MeatMask increased 0.001 ETH, which is the amount of all the Tips.



We can check the details of this operation at sepolia.etherscan.io

